

Pramesh Karmcharya

Senior Software Engineer

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Summary

Full Stack Developer with 6+ years of experience in designing and delivering dynamic, scalable, and high-performance web applications. Skilled in Node.js, Go, Python, PHP, React, and GenAI technologies, with a proven ability to architect and integrate robust backend systems, intelligent AI-driven solutions, and intuitive frontends for seamless user experiences.

I leverage modern AI frameworks and large language models (LLMs) to build adaptive, user-centric applications that combine innovation with practical functionality. Beyond technical expertise, I bring strong collaboration, problem-solving, and communication skills that enhance team efficiency and project success.

A commitment to continuous learning and innovation ensures I stay ahead of emerging technologies, making me a versatile and forward-thinking contributor to any project or team.

Technical Skills

Languages and Scripts: Go, Javascript, Typescript, Node.js, Python, Php, HTML5, CSS3

Frameworks and Libraries: ExpressJs, NestJs, Laravel, ReactJs, NextJs, Zustand, Tailwind, GoFiber, Chi, Gin, Htmx, Templ, FastAPI, RAG, MCP

Database: PostgreSQL MongoDB, MariaDB, CockroachDb, DynamoDB, DocumentDB

VCS: Git, Github, Gitlab

AWS services: EC2, S3, RDS, Cloudwatch, Lambda, X-ray, ECS, APIGateway, ECR, Bedrock, Opensearch

Containerization and orchestration: Docker

Collaboration: JIRA, Slack

Methodology: Scrum

Professional Experience

Senior Software Engineer (May 2025 - current) | Leapfrog Technology | Charkhal, Kathmandu

- Directed project architecture and technical strategy, mentoring junior developers and enforcing best practices in scalability, performance, and maintainability.
- Designed and implemented backend systems using Python, Node.js, Go,, and frontends with Next.js; delivered high-quality APIs and integrated with DynamoDB and DocumentDB
- Led the development of RAG-based chatbots with conversational memory and spearheaded a Presales AI Agent leveraging AWS Strands.
- Implemented AWS serverless architecture for the MCP RAG Starter Kit, enabling scalable, cost-efficient AI-powered applications.
- Oversaw deployment of applications on AWS ECR, ECS, EC2, and Lambdas, ensuring smooth and reliable production releases.
- Partnered with the People Management Team to drive successful hiring initiatives and conducted company-wide AI development tool training, enabling developers and QA engineers to adopt AI in their workflows.
- Ensured code quality and security through code reviews, performance profiling, and security best practices against SQL injection, XSS, and other vulnerabilities.
- Collaborated with designers, QA engineers, and project managers to align technical solutions with business objectives and deliver impactful outcomes.
- Explored emerging technologies, optimized workflows, and fostered a culture of knowledge sharing and continuous learning across teams.

Senior Software Engineer (March 2019 - April 2025) | Ekbana Solutions | Jwagal, Lalitpur

- Led the technical direction of the project, providing guidance and mentorship to junior developers
- Designed the architecture of the application, selecting appropriate technologies and frameworks to ensure scalability and performance.
- Implemented backend logic using Node.js/PHP, including the development of APIs and integration with databases like Postgresql or MongoDB.
- Worked on multiple frameworks like NextJs, NestJs, Go, and GoFiber for backend and frontend applications
- Ensured code quality by writing clean, maintainable code and conducting regular code reviews
- Collaborated with designers, front-end developers, and project managers to deliver successful outcomes
- Kept updated with the latest technologies and shared knowledge with the team
- Optimized application performance by profiling code and implementing optimizations

- Implemented security best practices to protect against common threats like SQL injection and XSS attacks

Software Engineer (May 2017 - January 2019) | Crystal Infosys | Tikathali, Lalitpur

- Wrote clean, efficient, and maintainable PHP code according to project requirements and coding standards. Identified and fixed bugs in existing codebase, collaborating with senior developers to ensure smooth application performance
 - Write and maintain server-side code using Node.js and JavaScript.
 - Continuously enhanced my skills by learning new technologies, tools, and best practices in PHP and Node.js ecosystem
 - Utilized version control systems such as Git to manage codebase and track changes effectively
 - Worked closely with other team members, including designers and project managers, to deliver projects on time and within budget
 - Contribute to code reviews to ensure quality and maintainability
 - Provided technical support to end-users and address any technical issues that arise during development or in production environments
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Education

Lincoln University

Master of Business Administration(MBA), | IIMS College| Expected 09/2027 | Gairidhara, Kathmandu, Bagmati

Tribhuvan University

Bachelor In Computer Science and Information Technology(BSc CSIT), | Orchid International College | 05/2019 | Bijaychowk, Kathmandu, Bagmati

Major Projects

Travel AI Assistant

A GenAI-powered application that helps users effortlessly plan personalized travel itineraries. Leveraging RAG-based chatbots and intelligent tools such as internet search, the assistant generates detailed travel plans tailored to the user's preferences and sends the finalized itinerary directly to their email for a seamless travel experience.

Roles:

- Led the development of APIs using FastAPI and built responsive UI components with React.

- Implemented the RAG chatbot using Qdrant VectorDB, ensuring accurate and context-aware responses for itinerary planning.
- Integrated internet search functionality via the Tavily MCP server to enrich travel recommendations with real-time information.
- Implemented AI workflows to combine LLM-generated insights with RAG outputs for highly accurate and user-specific travel itineraries.
- Designed logic to automate itinerary generation, combining AI insights, vector database retrieval, and real-time search data.
- Leveraged AWS LLM models, including Amazon Nova Pro and Claude Sonnet, through Bedrock to enhance AI-driven itinerary generation and conversational capabilities.
- Implemented validation, error handling, and logging to maintain reliability and user trust.

Presa Presales Agent

Presa Presales Agent is an AI-powered application that helps Account Managers, Project Managers, and Tech Leads streamline presales activities. It automates requirement analysis, competitive research, architecture discovery, user story mapping, and timeline estimations. Built with AWS Strands, Tavily, and Claude Sonnet 4, it provides intelligent, data-driven insights to improve presales efficiency and decision-making.

Roles:

- Led the end-to-end application development, including backend APIs and frontend components, ensuring robust, scalable, and user-friendly solutions.
- Designed and implemented the memory module using Mem0, enabling the application to retain context across multiple presales interactions for consistent and accurate outputs.
- Developed the authentication mechanisms to ensure secure and role-based access to sensitive presales data.
- Integrated AWS Strands, Tavily, and Claude Sonnet 4 to drive intelligent presales recommendations, architecture suggestions, and estimations.
- Worked closely with stakeholders to align AI-driven workflows with business needs, optimizing the application for performance, reliability, and usability.

Nepal Electricity Authority (NEA)

The application provides a platform to check the monthly electricity bills and pay the bills seamlessly within the app. It also allows the end users to view the electricity consumption on a daily, weekly, and monthly basis.

Roles:

- Created the API services and CMS using ExpressJs
- Worked on notifying the latest bills to end customers through Firebase notification using Bull MQ

- Worked on the offline bills data processing by creating the service in GO for concurrency of around 50 million records
- Worked on OTP information delivery and verification.

Pickelzone

PickleZone offers pickleball players a comprehensive platform to track game scores and organize teams for both singles and doubles matches. With intuitive score-tracking features, players can easily record points and monitor the progress of their games in real time. Whether it's a best-of-three or best-of-five series, PickleZone facilitates seamless organization, allowing players to create and join teams, set up matches, and compete with friends or rivals

Role:

- Designed and worked on the API service architecture in the CQRS pattern
- Designed and developed the CMS portal using React JS
- Used NestJS Monorepo Architecture for the multiple services
- Performed the unit testing using the Jest plugin for the core business logic
- Performed the E2E testing on the CMS services
- Performed the Unit testing on the API services
- Worked on RabbitMQ for the firebase push notifications and activity logging purposes and analytics
- Worked on the geofencing functions of PostgreSQL to track the location of players within the court radius
- Worked on S3 bucket to store the images and video files
- Worked on Dockerization on the Application.
- Deployed the application on AWS EC2 instance.

Salvi Task Management System

A web-based application that tracks the issues that occur in the street lamps. Basically, all the lamps are IoT devices, so when any issue occurs it publishes the issue through the MQTT protocol to the system. The system operator visualizes the issue in the system dashboard in real time and assigns the task to the associated technicians. Technicians receive the notification in the mobile App and look into the issue by accepting the task and completes it within the given deadline.

Role:

- Designed the overall system architecture using the Learn Monorepo for the multiple services for code reusability and maintainability
- Implemented the SSO (Single SignOn) using OpenId Connect protocol through KeyCloak Identity Provider
- Worked and implemented the MQTT protocol to interact with the devices and other services in real-time
- Worked on the Firebase push notification for mobile devices
- Worked on the frontend services on React JS

- Deployed the application using Pm2

Goth Starter

Goth Starter is a web-based application that is completely made on the Go ecosystem which provides a CMS platform in order to manage the content in the application

Role:

- Designed the overall codebase and architecture of the system
- Used the Go standard library and the Chi framework for routing
- Similarly used Gorm in order to perform the interactions with the database
- Used technologies like HTMX, and Templ library for template rendering
- Implemented the whole authentication and role-based permission system in Go
- Implemented the localization mechanisms from the standard Go i18n library
- Wrote the unit testing for the proper working of the services and modules
- Used Tailwind and DaisyUI for the designing and theming

My Place

MyPlace is a powerful web-based application that simplifies the management of real estate holdings. With MyPlace, users can easily keep track of properties, leases, tenants, and payments all in one place.

Role:

- Worked on the EUC module to manage the property data and generate reports in folders using Vue JS within the application
- Worked on the Postgresql Views and the materialized view for data management
- Worked on the Google Drive API to maintain the documents as per the property in the drive space
- Worked on the third-party APIs eg: Rakumo

Yeti Airlines

With this app, users can easily book flights, manage reservations, and access important travel information such as flight schedules and check-in details. Additionally, the app offers a range of helpful features such as real-time flight tracking and in-flight entertainment options to ensure a comfortable and stress-free journey.

Role:

- Worked on the API service and CMS
- Worked on the Core Flight Management SOAP API (Avantik Flight Management API)
- Worked on the Social Login (Google, Facebook, and Apple login)
- Worked on the Laravel Google Analytics SDK

IFA

IFA is a web-based application that assists financial advisors in creating and managing portfolios for their end customers. So the end customers can get the benefit from their investments.

Role:

- Worked on the Single Sign On (SSO) using One login for authentication to core existing core application
- Worked on Azure Active Directory for user management and authorization
- Used Redis to cache the APIsWorked on the data visualization part on charts and graphs of the portfolio data